

BRIAN M SCHILDER, PHD

Passionately pursuing transdisciplinary research to advance human health and knowledge.

STANDARD MODEL BIO

Genomic AI Scientist

Below are selected subsets of the full CV. -

EDUCATION

2024

Imperial College London / The Alan Turing Institute
PhD; Computational Genomics & Machine Learning — London, UK

Thesis: Multi-omic medicine: dissecting the cell type-specific and pleiotropic mechanisms underlying disease genomics at scale

2017



The George Washington University / Georgetown University
MPhil; Comparative Neuroscience & Genomics — Washington, DC, USA

Thesis: The evolution of the hippocampus and adult neurogenesis: novel insights into the origins of human memory

2011

Brown University / Princeton University
ScB; Neurological Diseases & Disorders — Providence, RI, USA

CORE SKILLS

Research

17+ years of deep expertise in genomics, AI, evolutionary biology and biomedicine. Strategically fuses concepts and methods across multiple domains.

- **Publication record**: 32 publications, 3 preprints and 14 awarded grants.
- **Reproducibility**: Global leader in promoting and enabling reproducible scientific practices. Writes 100% reproducible manuscripts programmatically.
- **Bioinformatics**: Created 50 Python and R packages to address key challenges in biological research.
- **High-performance computing**: Highly parallelised analyses and AI model training (CPUs and GPUs).
- **Web development**: 7+ websites, web apps, and interactive reports.

CONTACT

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LinkedIn
ORCID
Google Scholar
GitHub
Twitter
YouTube
Personal Website
Lab Website

SUMMARY

17+ years of research
32 publications
3 preprints
45 software packages
12 databases & apps
24 talks
15+ years of teaching & team management

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Education
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AI & Machine Learning

Proficient in developing and deploying AI/ML models (PyTorch, Keras, sklearn) to solve complex biological problems. Applied examples include:

- **Personalized VEP**: Adapted protein (ESM, AlphaFold), RNA (SpliceAI), and DNA (Flashzoi, phyloGPN, HyenaDNA) sequence models to generate personalized variant effect predictions for >100k clinical variants across >3,800 diverse human genomes. Demonstrated that sequence-based AI models capture biologically meaningful inter-individual variation, paving the way for personalized medicine.
- **TimelineTransformer**: Designed and trained an LLM-based time-series foundation model to predict patient health trajectories. Trained on 6.22M patient events.
- **Phenotype Severity Scoring**: Developed GPT-4-based framework to systematically score severity of 17k+ disease phenotypes to help reprioritization.
- **MedAgent**: A reproducible multi-agent system for summarizing and interpreting multimodal medical data, including EHR, imaging, and genomics.
- **ReporterSuite**: Created proprietary Python packages for advanced LLM-based topic modelling of the PubMed literature to provide business intelligence to the world's largest digital health, biotech, and pharma companies (as a consultant with 120/80 Group).
- **echolocatoR**: Used functional impact predictions from DNA sequence models (DeepSEA, Basenji, IMPACT) to validate SNPs prioritised with Bayesian fine-mapping.

Project Management

Efficient management strategies to define objectives, track progress and coordinate diverse teams.

- **Documentation**: Defines objectives and tracks progress with GitHub Projects. Includes useful documentation in Issues, inline code and shareable reports.
- **Version control**: Extensive and daily use of GitHub, containers (*Docker, Singularity, virtual machines*), environments (*conda*) and pipelines (*Nextflow*).
- **Team management**: Led numerous collaborative research projects and supervised researchers at various career stages.

Soft Skills

Advances science through effective problem formulation, collaboration and communication.

- **Problem formulation**: Rapid hypothesis generation, project design, and creative problem solving.
- **Collaboration**: Diverse and global collaborative networking.
- **Communication**: Clear and concise distillation of complex results to a variety of audiences. Presented **25** conference posters.

PUBLICATIONS

2026 **Cell type-specific contextualisation of the human phenome: towards the systematic treatment of all rare diseases**
Genome Medicine (2026) 18(113); https://doi.org/10.1186/s13073-026-01692-0
BM Schilder, KB Murphy, H Dash, Y Zhang, R Gordon-Smith, J Chapman, M Otani, NG Skene

2026 **anndataR improves interoperability between R and Python in single-cell transcriptomics**
Bioinformatics (2026) 42(6); https://doi.org/10.1093/bioinformatics/btag288
L Deconinck, L Zappia, R Cannoodt, M Morgan, scverse core, I Virshup, C Sang-Aram, D Bredikhin, **BM Schilder**, R Seurinck, Y Saeys

2026 **Using GPT-4 to annotate the severity of all phenotypic abnormalities within the human phenotype ontology**
Frontiers in Digital Health (2026) 8(1794934); https://doi.org/10.3389/fdgth.2026.1794934
KB Murphy, **BM Schilder**, NG Skene

2026 **Gene expression dynamics of human and mouse craniofacial development at the single-cell level**
Nature Communications (2026) 17(3714); https://doi.org/10.1038/s41467-026-70232-6
N Khouri-Farah, EW Winchester, **BM Schilder**, K Robinson, SW Curtis, NM Skene, E Leslie-Clarkson, J Cotney

2025 **monarchr: an R package for querying biomedical knowledge graphs**
Bioinformatics (2025) 41(10); https://doi.org/10.1093/bioinformatics/btaf549
ST O'Neil, **BM Schilder**, K Schaper, C Cox, D Korn, S Gehrke, CJ Mungall, MA Haendel

2025 **Mondo: integrating disease terminology across communities**
Genetics (2025) https://doi.org/10.1093/genetics/iyaf215
NA Vasilevsky, … **BM Schilder**, …,PN Robinson, CJ Mungall, A Hamosh, MA Haendel

2025 **Fine-mapping genomic loci refines bipolar disorder risk genes**
Nature Neuroscience (2025) https://doi.org/10.1038/s41593-025-01998-z
M Koromina, A Ravi, G Panagiotaropoulou, **BM Schilder**, … S Ripke, T Raj, JRI Coleman, N Mullins

2025 **CUT&Tag recovers up to half of ENCODE ChIP-seq peaks**
Nature Communications (2025) (16):2993; https://doi.org/10.1038/s41467-025-58137-2
L Abbasova, P Urbanaviciute, D Hu, JN Ismail, **BM Schilder**, A Nott, NG Skene, SJ Marzi

2025 **Chromatin Interaction and Histone Mark Signatures Associated With TBXT Expression in Metastatic Lung Cancer**
Genes Chromosomes Cancer, (2025) (64):e70041; https://doi.org/10.1002/gcc.70041
RM Yaa, **BM Schilder**, RD Acemel, FC Wardle

2023 **rworkflows: automating reproducible practices for the R community**
Nature Communications (2023) 15(149); https://doi.org/10.1038/s41467-023-44484-5
BM Schilder, AE Murphy, NG Skene
News
- Featured in *Nature Communications* Editors' Highlights

2023 **Artificial intelligence for neurodegenerative experimental models**
Alzheimer's & Dementia (2023) http://doi.org/10.1002/alz.13479
SJ Marzi, **BM Schilder**, A Nott, C Sala Frigerio, S Willaime-Morawek, M Bucholc, DP Hanger,C James, PA Lewis, I Lourida, W Noble, F Rodriguez-Algarra, JA Sharif, M Tsalenchuk, LM Winchester, U Yaman, Z Yao, DEMON Network, JM Ranson, DJ Llewellyn

2023 **Artificial intelligence for dementia genetics and omics**
Alzheimer's & Dementia (2023) http://doi.org/10.1002/alz.13427
C Bettencourt, NG Skene, S Bandres-Ciga, E Anderson, LM Winchester, IF Foote, J Schwartzentruber, JA Botia, M Nalls, A Singleton, **BM Schilder**, J Humphrey, SJ Marzi, CE Toomey, A Al Kleifat, EL Harshfield, V Garfield, C Sandor, S Keat, S Tamburin, C Sala Frigerio, I Lourida, DEMON Network, JM Ranson, DJ Llewellyn

2023 **Artificial intelligence for dementia research methods optimization**
Alzheimer's & Dementia (2023) http://doi.org/10.1002/alz.13441
M Bucholc, C James, A Al Khleifat, A Badhwar, N Clarke, A Dehsarvi, CR Madan, SJ Marzi, C Shand, **BM Schilder**, S Tamburin, HM Tantiangco, I Lourida, DJ Llewellyn, JM Ranson

2023 **EpiCompare: R package for the comparison and quality control of epigenomic peak files**
Bioinformatics Advances (2023) 13(1):vbad049; https://doi.org/10.1093/bioadv/vbad049
S Choi, **BM Schilder**, L Abbasova, AE Murphy, NG Skene

2022 **Dissecting the Shared Genetic Architecture of Suicide Attempt, Psychiatric Disorders, and Known Risk Factors**
Biological Psychiatry (2022) 91(3):313-327; https://doi.org/10.1016/j.biopsych.2021.05.029
N Mullins, J Kang, AI Campos,…**BM Schilder**, et al.

2022 **Genetic analysis of the human microglial transcriptome across brain regions, aging and disease pathologies**
Nature Genetics (2022) https://doi.org/10.1038/s41588-021-00976-y
K de Paiva Lopes, G JL Snijders, J Humphrey, A Allan, M Sneeboer, E Navarro, **BM Schilder**…T Raj
News
- Microglial transcriptomics meets genetics: new disease leads (Nature Reviews Neurology, 2022)
- Mighty MiGA: Microglial Genomic Atlas Zeros in on Causal AD Risk Variants (ALZFORUM, 2022)
- Can a Human Microglial Atlas Guide Brain Disorder Research? (Mount Sinai Health System, 2022)
- Polygenic Scores Paint Microglia as Culprits in Alzheimer's (ALZFORUM, 2021)

2021 **Multi-omic insights into Parkinson's Disease: From genetic associations to functional mechanisms**
Neurobiology of Disease (2021) 105580; https://doi.org/10.1016/j.nbd.2021.105580
BM Schilder, E Navarro, T Raj

2021 **Fine-Mapping of Parkinson's Disease Susceptibility Loci Identifies Putative Causal Variants**
Human Molecular Genetics (2021) ddab294; https://doi.org/10.1093/hmg/ddab294
BM Schilder, T Raj

2021 **echolocatoR: An Automated End-to-End Statistical and Functional Genomic Fine-Mapping Pipeline**
Bioinformatics (2021) btab658; https://doi.org/10.1093/bioinformatics/btab658
BM Schilder, J Humphrey, T Raj

2021 **MungeSumstats: A Bioconductor Package for the Standardisation and Quality Control of Many GWAS Summary Statistics**
Bioinformatics (2021) 37(23):4593-4596; https://doi.org/10.1093/bioinformatics/btab665
A Murphy, **BM Schilder**, NG Skene

2021 **Dysregulation of mitochondrial and proteo-lysosomal genes in Parkinson's disease myeloid cells**
Nature Genetics (2021) https://doi.org/10.1101/2020.07.20.212407
E Navarro, E Udine, K de Paiva Lopes, M Parks, G Riboldi, **BM Schilder**…T Raj
News
- Mount Sinai: Fighting Neurodegenerative Disorders (Mount Sinai Health System, 2019)

2021 **Phenome-wide and eQTL Associations of COVID-19 Genetic Risk Loci**
iScience (2021) https://doi.org/10.1016/j.isci.2021.102550
C Moon, **BM Schilder**, T Raj, K-I Huang

2021 **Genome-Wide Association Study of over 40,000 Bipolar Disorder Cases Provides Novel Biological Insights**
Nature Genetics (2021) 53:817-829; https://doi.org/10.1038/s41588-021-00857-4
N Mullins, AJ Forstner, KS O'Connell, B Coombes, JRI Coleman…**BM Schilder**… et al.
News
- Researchers identify 64 regions of the genome that increase risk for bipolar disorder (EurekAlert, 2021)
- Largest Bipolar Disorder Genetics Study Doubles Genetic Risk Factors (Nordic Society of Human Genetics and Precision Medicine, 2021)

2020 **Tensor decomposition of stimulated monocyte and macrophage gene expression profiles identifies neurodegenerative disease-specific trans-eQTLs**
PLOS Genetics (2020) 16(9):e1008549; https://doi.org/10.1371/journal.pgen.1008549
S Ramdhani, E Navarro, E Udine, AG Efthymiou, **BM Schilder**, M Parks, A Goate, T Raj

2019 **Evolutionary shifts dramatically reorganized the human hippocampal complex**
Journal of Comparative Neurology (2019) 528(17):3143-3170; https://doi.org/10.1002/cne.24822
BM Schilder, HM Petry, PR Hof

2019 **FAIRshake: Toolkit to Evaluate the Findability, Accessibility, Interoperability, and Reusability of Research Digital Resources**
Cell Systems (2019) 9; https://doi.org/10.1016/j.cels.2019.09.011
D Clarke, L Wang, A Jones, M Wojciechowicz, D Torre, K Jagodnik, S Jenkins, P McQuilton, Z Flamholz, M Silverstein, **BM Schilder**…A Ma'ayan
News
- Chosen as 'Featured Frontmatter' article in Cell Systems

2019 **Geneshot: search engine for ranking genes from arbitrary text queries**
Nucleic Acids Research (2019) 47(W1):W571-W577; https://doi.org/10.1093/nar/gkz393
A Lachmann, **BM Schilder**, ML Wojciechowicz, D Torre, MV Kuleshov, AB Keenan, A Ma'ayan
News
- Geneshot: Piercing the Literature to Identify and Predict Relevant Genes (University of Pittsburgh Health Sciences Library System Update, 2019)
- The Future of AI at the Hasso Plattner Institute for Digital Health at Mount Sinai (Mount Sinai Health System, 2020)

2018 **eXpression2Kinases (X2K) Web: linking expression signatures to upstream cell signaling networks**
Nucleic Acids Research (2018) 46(W1):W171-W179; https://doi.org/10.1093/nar/gky458
DJB Clarke, MV Kuleshov, **BM Schilder**, D Torre, ME Duffy, AB Keenan, A Lachmann, AS Feldmann, GW Gundersen, MC Silverstein, Z Wang
News
- Mount Sinai Faculty Spotlight: Ma'ayan Lab (Mount Sinai Health System, 2018)

2015 **Defining elemental imitation mechanisms: A comparison of cognitive and motor-spatial imitation learning across object- and computer-based tasks**
Journal of Cognition and Development (2015) 17(2):221-243; https://doi.org/10.1080/15248372.2015.1053483
F Subiaul, L Zimmerman, E Renner, **BM Schilder**, R Barr

2015 **Take the monkey and run**
Journal of Neuroscience Methods (2015) 248:28-31; http://doi.org/10.1016/j.jneumeth.2015.03.023
KA Phillips, MK Hambright, K Hewes, **BM Schilder**, CN Ross, SD Tardif
News
- Monkeys on a Treadmill? A Conversation with Dr. Kimberley Phillips (Why Social Science?)

2014 **Becoming a high-fidelity - super - imitator: what are the contributions of social and individual learning?**
Developmental Science (2014) 18(6):1025-1035; http://doi.org/10.1111/desc.12276
F Subiaul, EM Patterson, **BM Schilder**, E Renner, R Barr

2014 **Working memory constraints on imitation and emulation**
Journal of Experimental Child Psychology (2014) 128:190-200; http://doi.org/10.1016/j.jecp.2014.07.005
F Subiaul, **BM Schilder**

PREPRINTS

2026 **Genetic background shapes AI-predicted variant effects**
bioRxiv (2026) https://doi.org/10.64898/2026.04.04.715328
BM Schider, Z Liu, JJ Desmairis, D Laub, F Rahimi, P Sethi, LA Pereira, M Sun, JB Kinney, DM McCandlish, J Zhou, PK Koo

2026 **orthogene: a Bioconductor package to easily map genes within and across hundreds of species**
bioRxiv (2026) https://doi.org/10.64898/2026.01.17.700094
BM Schilder, AE Murphy, NG Skene

2024 **Integrative multi-omics analysis of glial signatures associated with accelerated cognitive decline in Alzheimer's disease**
bioRxiv (2024) https://doi.org/10.1101/2024.08.27.24312641
E Schneegans, N Fancy, V Chau, TKD Cheung, E Adair, M Papageorgopoulou, **BM Schilder**, PM Matthews, JS Jackson

RESEARCH EXPERIENCE

- | 2026 **AI Scientist**
Standard Model Biomedicine — Philadelphia, PA, USA
• Building multimodal world models of patient health. Specifically, I lead the design and development of AI models that transform whole-genomes into personalized clinical predictions.

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2024

Postdoctoral Research Scientist
Cold Spring Harbor Laboratory (Simons Center for Quantitative Biology) Cold Spring Harbor, NY, USA
- Advancing deep learning applications in genomics and biomedicine in the laboratory of Dr. Peter Koo.
- Developing a genomic foundation model to map complex genome-phenome relationships and make highly accurate, personalized disease risk predictions.

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2019

Lead Data Scientist
120/80 Group New York, NY, USA
- Offers data-driven consultation services to a wide portfolio of high-profile digital healthcare, pharmaceutical and biotech companies.
- Developed a suite of proprietary softwares to extract customised business intelligence from the published literature to generate customised and interpretable reports to clients.
- Provides clients guidance on strategic AI implementation, data analysis, publication and transparency.

2020
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2018

Bioinformatician II
Icahn School of Medicine at Mount Sinai (Department of Neuroscience / Department of Neurology / Department of Genetics & Genomics / Ronald M. Loeb Center for Alzheimer's Disease)
New York, NY, USA
- Developed machine learning systems to integrate large-scale multi-omics datasets (e.g. whole-genome sequencing, bulk and single-cell RNA-seq, epigenomics, clinical data) to uncover the molecular mechanisms underlying neurodegenerative diseases (e.g. Alzheimer's, Parkinson's, ALS).
- Computationally identified specific disease-causal variants, pathways and cell-types for subsequent functional wet lab validation (e.g. CRISPR-cas9 editing in patient-derived cell cultures, iPSCs and cerebral organoids).

2018
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2017

Bioinformatician II
Icahn School of Medicine at Mount Sinai (Department of Pharmacological Sciences) New York, NY, USA
- Conducted computational systems biology research. Integrated and analyzed large-scale genomic and biomedical data (e.g. Python, R, JavaScript).
- Developed evolutionary algorithm to optimize gene network kinase regulator prediction (eXpression2Kinases).
- Developed and deployed computational tools, software, databases and web applications for basic and clinical research, resulting in 3 peer-reviewed publications.

2013
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2011

Research Assistant
The George Washington University (Department of Anthropology) Washington, DC, USA
- Performed dissection, histology, microscopy and quantitative stereology in post-mortem primate brain tissues.
- Trained junior and senior personnel on lab protocols.

2013
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2011

Senior Lab Manager
The George Washington University (Department of Speech, Language & Hearing Sciences)
Washington, DC, USA
- Organized and trained dozens of undergraduates to conduct weekly cognitive development research; designed and/or directly contributed to over 15 research projects in two years.

2010

Paid Research Intern
Princeton University (Princeton Neuroscience Institute) Princeton, NJ, USA
- Investigated the neural basis of decision-making in humans.
- Recruited participants, recorded EEG and analyzed data in MATLAB.

$ GRANTS

Total (all grants): $3,049,872
Total (as primary applicant): $311,382